Practice Set 12 Paper 1 – Basic Sciences in Urology

1. Q1. Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children
2. Q2. Lymphatic drainage of testes and principles of designing templates of RPLND
3. Q3a. Retroperitoneal fibrosis
4. Q3b. TIP repair in hypospadias
5. Q3c. Perineal urethrostomy.
6. Q3d. Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options.
7. Q3e. Neurotransmitters in erectile function
8. Q3f. TUR Syndrome

Practice paper compiled from the question bank.

Q1. Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children

Type: LAQ (25 marks) (frequency and urgency in a child also in 2017-Summer Paper 3, Q3c)

Introduction

- The LUT (lower urinary tract) has two opposite functions: **storage** at low pressure with a closed outlet, and **emptying** by a coordinated detrusor contraction with a relaxed outlet. Bladder and sphincter act reciprocally.

Peripheral innervation

	Parasympathetic	Sympathetic	Somatic
Origin	**S2–S4** (sacral parasympathetic nucleus)	Thoracolumbar (T10–L2)	**Onuf nucleus** (S2–S4 ventral horn)
Nerve	Pelvic	Hypogastric	Pudendal
Bladder	Contracts (ACh [acetylcholine] → **M3**; ATP)	Relaxes (noradrenaline → **β3**)	–
Outlet	Relaxes urethra (**NO**, nitric oxide)	Contracts bladder neck (**α1**)	Contracts EUS (external urethral sphincter; nicotinic)
Role	Voiding	Storage	Voluntary closure

- All three nerves carry afferents. **Aδ fibres** (tension receptors in muscle) signal normal fullness. **Unmyelinated C fibres** are "silent" normally; after inflammation, SCI (spinal cord injury) or obstruction they are recruited and drive overactivity (basis of capsaicin and resiniferatoxin).
- Detrusor has more M2 receptors, but **M3 mediates contraction**.

Reflex and central control

- **Storage (spinal reflexes):** low afferent activity → sympathetic outflow (detrusor inhibition, bladder-neck closure) and the **guarding reflex** (pudendal → rising sphincter activity with filling).
- **Voiding (spinobulbospinal reflex):** high afferent activity → **PAG (periaqueductal grey)** → **PMC (pontine micturition centre, Barrington nucleus)** → sphincter relaxes first, then detrusor contracts. PMC is an on–off switch.
- **Forebrain decides:** insula senses fullness; **medial prefrontal cortex** tonically holds the switch "off" until voiding is socially appropriate.
- Lesion patterns: above pons → overactivity, coordinated sphincter; between pons and sacral cord → overactivity with **DSD (detrusor–sphincter dyssynergia)**; sacral/peripheral → underactive bladder.
- **Central transmitters:** glutamate excitatory; GABA (γ-aminobutyric acid) and glycine inhibitory; serotonin inhibits voiding; dopamine tonically inhibits the PMC (lost in Parkinson disease).

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Filling → Aδ afferents → sacral cord → PAG
  low traffic: spinal storage reflexes (sympathetic + guarding)
  high traffic + cortex "permits": PMC → EUS relaxes → detrusor contracts
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Maturation in children

- Bladder visible on fetal ultrasound by 14 weeks. Infants void reflexly but **wake before voiding** (cortical arousal gates voiding).

- Infant voiding is often interrupted (transient dyscoordination) with detrusor pressure often **>100 cmH2O**; frequent small voids largely disappear by **2 years**.
- Capacity rises: **EBC (expected bladder capacity) = (age + 2) × 30 mL** (infants: weight in kg × 7 mL).
- Toilet training: bowel control → daytime bladder → night-time bladder. Persisting dyscoordination after toddlerhood is dysfunction, not immaturity.

Types of voiding dysfunction in children

A. Non-neurogenic (functional) LUT dysfunction (ICCS [International Children's Continence Society])

Type	Key feature
OAB (overactive bladder)	Urgency ± frequency (≥8 voids a day), urgency incontinence; tower flow
Voiding postponement	Habitual holding; holding manoeuvres; infrequent voids (≤3 a day)
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction during voiding in a neurologically normal child; **staccato** flow
Underactive bladder	Needs abdominal straining to void; interrupted flow
Giggle incontinence	Complete emptying with laughter
Vaginal reflux	Small leak just after voiding in girls
Enuresis	Wetting in sleep, ≥5 years; monosymptomatic or non-monosymptomatic

- **BBD (bladder and bowel dysfunction):** LUT dysfunction plus constipation; rectal loading worsens symptoms and UTI (urinary tract infection).
- **Hinman (non-neurogenic neurogenic bladder):** severe end of dysfunctional voiding → trabeculated bladder, VUR (vesicoureteric reflux), renal damage.

B. Neurogenic LUT dysfunction

- Causes: **myelomeningocele** (commonest), occult dysraphism/**tethered cord**, sacral agenesis, anorectal malformation, cerebral palsy, SCI, transverse myelitis, pelvic tumour surgery (sacrococcygeal teratoma).
- Pattern = detrusor (overactive or underactive) × outlet (competent, incompetent or dyssynergic). Bony level does not predict bladder function.
- **Hostile bladder: DLPP (detrusor leak point pressure) ≥40 cmH2O** (McGuire 1981), poor compliance or NDO (neurogenic detrusor overactivity) with DSD.

C. Anatomical: PUV (posterior urethral valves) "valve bladder": overactivity → poor compliance → myogenic failure; ectopic ureter.

Evaluation

- Bladder and bowel diary, Bristol stool chart, spine and neurological examination, urinalysis.
- **Uroflowmetry with PVR (post-void residual) and patch EMG (electromyography)**: bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive/DSD), plateau (obstruction).
- Invasive (video)urodynamics only for neurogenic, anatomical or refractory cases.

Management principles

- **Treat constipation first**; standard urotherapy (timed voiding, posture, fluids) for 4 weeks.
- Dysfunctional voiding: **biofeedback**; α-blocker; sphincter botulinum toxin if refractory.
  OAB: **oxybutynin 0.3–0.6 mg/kg/day** (maximum 15 mg); mirabegron; botulinum toxin; parasacral TENS (transcutaneous electrical nerve stimulation).

- Enuresis: alarm or desmopressin.
- Neurogenic: **CIC (clean intermittent catheterisation; Lapides 1971) + antimuscarinic** → intradetrusor botulinum toxin → augmentation.

Diagram to draw: bladder with pelvic, hypogastric and pudendal nerves, plus brainstem loop (PAG–PMC).

> **Key points to remember:**
> - S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
> - PMC = switch; medial prefrontal cortex holds it off.
> - EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
> - DLPP ≥40 cmH2O = hostile; treat bowel first.

Q2. Lymphatic drainage of testes and principles of designing templates of RPLND

Type: LAQ (25 marks) (same question as Practice Set 7 Paper 1, Q2; testicular lymphatics also in 2018-Summer Paper 2, Q2; MI-RPLND in 2025-Winter Paper 3, Q3f)

Lymphatic drainage of the testis

- **Why retroperitoneal:** the testis develops near the kidney (L1–L2), so its artery, vein and lymphatics stay retroperitoneal.
- **Route:** lymphatics run in the cord with the testicular vessels through the internal ring, then along the gonadal vessels to lumbar nodes near the renal hilum. Lymph capillaries do not enter seminiferous tubules.
- Onward: lumbar trunks → **cisterna chyli** (behind the right aorta, in front of L1–L2, behind the right crus; hence retrocrural nodes) → **thoracic duct** → left innominate vein. Explains **left supraclavicular** metastases.

Retroperitoneal nodal regions

Region	Extent
Right paracaval (with precaval, retrocaval)	IVC (inferior vena cava) midline to right ureter
Interaortocaval	IVC midline to aortic midline
Left para-aortic (with pre-aortic, retroaortic)	Aortic midline to left ureter

Landing zones

	Right testis	Left testis
Primary	**Interaortocaval** (below renal vessels)	**Left para-aortic** and left hilar
Next echelon	Paracaval, precaval, then para-aortic	Interaortocaval
Contralateral	Common	Rare; implies bulk

- **Crossover is right to left** because lymph drifts right to left.
- **Epididymis:** caput and corpus with the testis; **cauda** with the vas to **external iliac** nodes.
- **Iliac and inguinal nodes** only with retrograde spread from bulky disease, or altered drainage after scrotal violation, inguinal or scrotal surgery, or tumour reaching the tunica/scrotum.
- **Scrotum:** ipsilateral superficial inguinal nodes; nothing crosses the raphe. Hence never biopsy or remove a testis trans-scrotally.
- The retroperitoneum is the first metastatic site in 70–80% (choriocarcinoma spreads by blood). CT (computed tomography) understages 25–35% of stage I NSGCT (non-seminomatous germ cell tumour); 5–9 mm landing-zone nodes are suspicious.

Principles of designing RPLND (retroperitoneal lymph node dissection) templates

1. Map-based design (mapping studies)

- Historical dissection: bilateral infrahilar + suprahilar + interiliac: most men anejaculatory.
- **Ray (1974)** first templated landing zones; **Donohue (1982)** mapping confirmed side-specific spread; **Weissbach and Boedefeld (1987)** reduced left template with frozen section.
- So a template must include the primary landing zone, the next echelon and the known crossover route.

2. Fixed boundaries

- **Superior:** renal hila (renal vessels). **Lateral:** ureters. **Inferior:** ureter crossing the common iliac artery. Plus the **gonadal vein and cord stump** to the internal ring.
- Omit **suprahilar** (less chylous ascites, renal, pancreatic and duodenal injury) and **interiliac** regions in non-bulky disease, as spread there is rare.

3. Templates

Template	Content
Full bilateral infrahilar	Paracaval, precaval, retrocaval, interaortocaval, pre-aortic, para-aortic, retroaortic, both common iliacs
Modified right	Right paracaval, pre- and retrocaval, interaortocaval, right common iliac; para-aortic **above the IMA (inferior mesenteric artery)**; omit para-aortic below IMA
Modified left	Para-aortic, pre- and retroaortic, left common iliac, interaortocaval above IMA; omit paracaval/precaval/retrocaval

- Omitting para-aortic above the IMA (right) or interaortocaval (left) is controversial.

4. Preserve ejaculation

- Emission and bladder-neck closure depend on **L1–L4 postganglionic sympathetic fibres** (about three each side) joining the **superior hypogastric plexus at or below the IMA** and running over the sacral promontory.
- **Unilateral templates** leave contralateral fibres untouched (antegrade ejaculation 75–97%). **Prospective nerve sparing** (Jewett 1988) identifies fibres: 90–100%.

5. Completeness ("duplicate open RPLND")

- **Split-and-roll** over the great vessels; **divide lumbar vessels** to clear retroaortic and retrocaval tissue; clip lymphatics (chylous leak). Incomplete RPLND or mass excision gives worse outcomes (GETUG T93BP; 33% local relapse after post-chemotherapy mass-only resection).

6. Setting decides extent

	Primary RPLND	PC-RPLND (post-chemotherapy)
Default	Modified unilateral or full bilateral nerve-sparing (no randomised trial)	Full bilateral infrahilar + residual mass
Modified allowed	Low-volume, clinically negative nodes	Only if mass ≤5 cm in landing zone, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good/intermediate risk
Main risk	Out-of-template nodes 3–23% (right > left)	Crossover with bulk; viable tissue unrecognisable

- Convert to bilateral if nodes are suspicious at surgery or the primary has somatic-type malignancy.
- *Sources differ:* Campbell testis-neoplasm chapter calls full bilateral nerve-sparing the standard; the surgery chapter accepts either.

Outcomes and complications

- Primary RPLND: pN+ about 20–30%; CSS (cancer-specific survival) 98.8–100%; retroperitoneal recurrence <2% after full template.
- Complications: ileus (commonest), wound infection, chylous ascites (0.2–2.1% primary; 2–12% post-chemotherapy), pulmonary (bleomycin), ejaculatory loss.

Also know: robotic RPLND uses the same templates; RPLND for small-volume stage II seminoma (SEMS, PRIMETEST).

Diagram to draw: retroperitoneum showing right and left modified templates, IMA level and hypogastric plexus.

Key points to remember:

- Right testis → interaortocaval; left → para-aortic; crossover right to left only.
- Boundaries: renal hila, ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve fibres L1–L4 meet at the IMA: unilateral templates spare ejaculation.
- Post-chemotherapy: full bilateral template remains standard.

Q3a. Retroperitoneal fibrosis

Type: SAQ (10 marks) (also answered as an LAQ in 2023-Winter Paper 3, Q1)

Definition

- RPF (retroperitoneal fibrosis): inflammatory fibrous plaque encasing the aorta, IVC (inferior vena cava) and ureters; centred on the **distal aorta at L4–L5**. Age 40–60; males 2–3 times commoner.

Aetiology

Idiopathic (~70%)	Secondary (~30%)
Chronic periaortitis (large-vessel vasculitis); **IgG4 (immunoglobulin G4)-related disease**; autoimmune; ceroid antigen	**Drugs** (methysergide, ergots, β-blockers); **malignancy** (lymphoma commonest); radiotherapy; infection (tuberculosis)

Pathology

- Early: lymphocytes, plasma cells, capillaries (steroid-responsive). Late: acellular collagen.
- Ureteric obstruction is by compression and **loss of peristalsis**; lumen stays patent. Malignant RPF shows tumour islands: take deep biopsies.

Clinical features

- **Dull back or flank pain**, weight loss, malaise, hypertension; about half present with **uraemia**; leg oedema or DVT (deep venous thrombosis) from IVC compression.

Evaluation

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein), creatinine, IgG4.
- **CT (computed tomography):** periaortic mass with **medial deviation of mid-ureters**.
- **MRI:** high T2 = active. **PET-CT (positron emission tomography):** best marker of activity.
- **Biopsy** if atypical or malignancy suspected.

Management

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Drain (stents or nephrostomy) → stop drug, exclude cancer
  → prednisone → failure / fibrotic disease → ureterolysis
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- **Prednisone** 60 mg daily (~1 mg/kg), tapered over up to 2 years: ~80% respond; relapse 25–50%.
- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% prednisone vs 50% tamoxifen.
- Steroid-sparing: azathioprine, mycophenolate; **rituximab** for refractory or IgG4 disease.
- **Ureterolysis** (open, laparoscopic or robotic; ~15% need it): **bilateral**, biopsies, then **intraperitonealisation**, lateralisation or **omental wrap**.
- Others: long-term metallic stents; autotransplantation; nephrectomy for a dead kidney.

Follow-up

- ESR/CRP, creatinine and ultrasound; MRI or PET-CT for activity; remove stents once the ureters drain; watch for relapse and steroid side effects.

Diagram to draw: periaortic plaque pulling both ureters medially.

Key points to remember:

- 70% idiopathic; methysergide and lymphoma classic.
- Medially deviated ureters; PET-CT shows activity.
- Prednisone first (Vaglio 2011 beat tamoxifen).
- Bilateral ureterolysis with omental wrap.

Q3b. TIP repair in hypospadias

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 3, Q3c; extended indications in Practice Set 5 Paper 3, Q3c)

Introduction

- **TIP (tubularised incised plate) urethroplasty**, Snodgrass (1994), built on Rich's hinging incision (1989).
- Used for **>90% of distal** and **about 80% of midshaft** repairs.

Principle

- A **deep midline relaxing incision** widens a narrow urethral plate so it can be tubularised **without tension**.
- The raw incised area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

Indications and prerequisites

- Distal and midshaft hypospadias; redo cases with a healthy plate; curvature **<30°** corrected by degloving or dorsal plication.
- **Glans width <14 mm** raises complications: consider preoperative testosterone.
- **POST (plate objective scoring tool):** a poor plate favours a dorsal inlay graft.
- Not for curvature needing plate division; proximal TIP largely abandoned.

Operative steps

1. Glans traction suture; tourniquet.
2. U-incision around plate and meatus; deglove; artificial erection.
3. Mobilise glans wings deeply off the plate.
4. **Midline plate incision** from meatus to plate end, down to corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, distal to proximal, leaving an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier layer.
7. Two-layer glansplasty; mucosal collar; Byars flaps or preputioplasty.
8. Stent 5–7 days; dressing; oxybutynin for bladder spasm.

Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, **meatal stenosis 3.6%**, stricture 1.3%.
- Glans dehiscence; meatal stenosis commoner than after Mathieu.
- Proximal and redo cases do worse; a dartos layer reduces fistula.

Modifications

- **Dorsal inlay graft (Snodgraft)**: inner preputial graft in the incised bed for a shallow plate.

Diagram to draw: cross-sections of plate incision, tubularisation and dartos cover.

> **Key points to remember:**
> - Snodgrass 1994: incise, tubularise, heals by epithelialisation.
> - Glans <14 mm and poor plate raise complications.
> - Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Q3c. Perineal urethrostomy.

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 2, Q3c)

Definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses**, rather than reconstructs, a diseased anterior urethra.

Indications

- **Complex anterior strictures:** LS (lichen sclerosus) panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk men** unwilling or unfit for staged reconstruction.
- **First stage** of staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for cancer.
- Temporary: after phalloplasty strictures; Fournier gangrene; resectoscope access in morbid obesity.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU suits LS.

Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS raise restricture.

Techniques

1. Classical (Blandy) PU: inverted-U perineal skin flap advanced into a ventral bulbar urethrotomy.

2. BMG (buccal mucosal graft) PU, steps:

1. High lithotomy; inverted-U incision; split bulbospongiosus in the midline.
2. Free and **transect** the spongiosum; **dorsal urethrotomy** into healthy mucosa.
3. Bring healthy ventral urethra to skin; calibrate proximally to 30 Fr; close distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin.

- The graft prevents the circumferential scar that narrows a deep PU.

Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Compared with staged reconstruction it is a shorter, single operation with less morbidity.
- High mid-term success and satisfaction in selected patients.

Diagram to draw: sagittal perineum with the inverted-U flap sutured into the opened bulbar urethra.

> **Key points to remember:**
>
> - PU bypasses the diseased anterior urethra; patient voids sitting.
> - Ideal for LS, failed repairs and elderly men.
> - BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Q3d. Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options.

Type: SAQ (10 marks) *Assumed meaning: pelvic organ prolapse (POP) in women.* (pelvic floor anatomy also in Practice Set 2 Paper 1, Q1)

Definition

- POP (pelvic organ prolapse): descent of the anterior wall, posterior wall, uterus/cervix or vault. Anterior > posterior > apical. It is a disease only if symptomatic.
- Cause: loss of **DeLancey support** (level I uterosacral–cardinal apex; II paravaginal to the ATFP [arcus tendineus fasciae pelvis]; III perineal body) and levator ani injury. Risks: vaginal delivery, parity, age, obesity.

Clinical presentation

- **Vaginal bulge** (most specific), pressure, heaviness, worse by evening.
- **Splinting/digitation** to void or defecate (high-grade POP).
- Urinary: voiding difficulty (urethral kinking), OAB (overactive bladder); SUI (stress urinary incontinence) in ~40%, may be **occult**.
- Bowel (rectocele): obstructed defecation, stool trapping. Sexual dysfunction, discharge, ulceration; procidentia may cause hydronephrosis.

Diagnostic evaluation

- Examine supine and standing on strain; split speculum per compartment.
- **POP-Q (Pelvic Organ Prolapse Quantification, 1996):** points Aa, Ba, C, D, Ap, Bp measured from the hymen (above negative), plus gh, pb, tvl.

Stage	Leading edge
0	No prolapse
I	< −1 cm
II	−1 to +1 cm
III	> +1 cm but < (tvl − 2) cm
IV	Complete eversion

- Urinalysis, PVR (post-void residual), diary; **cough test or urodynamics with prolapse reduced** for occult SUI; renal ultrasound (procidentia); dynamic MRI or translabial ultrasound if complex or recurrent; pessary trial confirms POP causes the symptoms.

Surgical options

Type	Operations
Restorative (native tissue)	Anterior colporrhaphy; paravaginal repair; posterior midline plication; perineorrhaphy
Apical (vaginal)	**Uterosacral suspension**, **sacrospinous fixation** (OPTIMAL: similar), vaginal hysterectomy; uterus-sparing hysteropexy, Manchester
Compensatory (abdominal)	**Sacrocolpopexy** (open, laparoscopic, robotic): most durable; mesh exposure ~3–10%
Obliterative	**Le Fort / total colpocleisis**: elderly, not sexually active

- Transvaginal synthetic mesh for POP is withdrawn (FDA order, 2019). Add a sling for occult SUI (OPUS, CARE trials).

Diagram to draw: DeLancey three levels of support.

Key points to remember:

- Bulge is most specific; splinting means high grade.
- POP-Q stage II = −1 to +1 cm.
- Sacrocolpopexy most durable; colpocleisis for elderly.

Q3e. Neurotransmitters in erectile function

Type: SAQ (10 marks) (erection physiology also in Practice Set 5 Paper 1, Q2)

Introduction

- Erection is **cavernous smooth-muscle relaxation**: low cytosolic Ca^{2+} dilates helicine arteries and sinusoids; expanding sinusoids compress emissary veins (veno-occlusion).
- Pathways: parasympathetic **S2–S4** cavernous nerves (erection), sympathetic **T11–L2** (detumescence), pudendal (sensation, ischiocavernosus rigidity).

Peripheral pro-erectile transmitters

Transmitter	Source / mechanism
NO (nitric oxide)	Principal. **nNOS (neuronal NOS)** initiates; **eNOS (endothelial)** sustains. NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → PKG (protein kinase G) → K^+ channels open, Ca^{2+} channels close → relaxation
ACh (acetylcholine)	Indirect: endothelial NO release, inhibits adrenergic nerves; atropine does not block erection
PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide), CGRP (calcitonin gene-related peptide)	**cAMP (cyclic adenosine monophosphate)** → PKA (protein kinase A); VIP role minimal in man
CO, H2S, EDHF (endothelium-derived hyperpolarising factor)	Gaseous/endothelial relaxants via cGMP or K^+ channels

- **PDE5 (phosphodiesterase 5)** breaks down cGMP and ends erection; PDE5 inhibitors block it.
- O2 is a NOS substrate: low flaccid Po2 (~35 mmHg) limits NO; erect Po2 ~90 mmHg.

Anti-erectile (flaccidity and detumescence)

- **Noradrenaline (α1)**: principal; α2 prejunctional reduces noradrenaline and NO release.
- **Endothelin-1** (ET-A), angiotensin II (AT1), PGF2α, thromboxane A2: Gq → IP3 → Ca^{2+} rise → Ca^{2+}–calmodulin → MLC (myosin light-chain) kinase → contraction.
- **Rho-kinase** inhibits MLC phosphatase (calcium sensitisation), sustaining tone.

Central neurotransmitters (PVN [paraventricular nucleus], MPOA [medial preoptic area])

Promote	Inhibit
Dopamine (D2 in PVN; apomorphine)	**Serotonin** (5-HT1A, 5-HT2; but 5-HT2C pro-erectile)
Oxytocin	GABA (γ-aminobutyric acid)
Central NO	Opioids (high dose), cannabinoids
Central noradrenaline; melanocortins (MC4R)	**Prolactin**

- Dopamine and oxytocin converge on NO in the PVN. Noradrenaline is pro-sexual centrally but anti-erectile peripherally.

Clinical relevance

- PDE5 inhibitors, intracavernosal alprostadil (cAMP), papaverine, phenylephrine for priapism (α1), SSRIs (selective serotonin reuptake inhibitors) cause dysfunction, hyperprolactinaemia lowers libido.

Diagram to draw: NO–cGMP pathway in the smooth-muscle cell.

Key points to remember:

- nNOS initiates, eNOS maintains; PDE5 terminates.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.

Q3f. TUR Syndrome

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 3, Q3f)

Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorbing hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). Incidence low and falling.

Pathophysiology

- Prostatic venous pressure about **10 mmHg**; irrigant at higher pressure enters open sinuses or, after perforation, the periprostatic space.
- Monopolar needs ion-free fluid: **glycine 1.5%**, sorbitol, mannitol or water (all hypo-osmolar).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water enters brain → **cerebral oedema**.
- **Glycine toxicity:** ammonia → encephalopathy; retinal inhibitory transmitter → **transient blindness**.

Risk factors

- Gland **>45 mL**, resection **>90 minutes** (Veterans Affairs Cooperative Study), high irrigation pressure, capsular perforation, sterile water.

Clinical features

- Awake (spinal) patient: restlessness, confusion, **nausea, vomiting, visual disturbance**.
- **Hypertension with bradycardia**, tachypnoea, later hypotension.
- Severe: seizures, coma, herniation. **Na <120 mEq/L** is dangerous.

Prevention

- **Bipolar TURP or laser with saline** (NNT [number needed to treat] about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous flow; low pressure.
- Limit resection time; stage huge glands; check sodium in long cases.
- Spinal anaesthesia for early detection; ethanol-tagged irrigant monitors absorption.

Treatment

1. Stop surgery, secure haemostasis, insert catheter.
2. ABC (airway, breathing, circulation), oxygen, monitoring; send **sodium**, osmolality, glucose.
3. Overload: **furosemide**; restrict fluids.
4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes, maximum 300 mL. **Target rise 4–6 mEq/L.**
5. Limit: **≤12 mEq/L in 24 h** for acute post-TURP hyponatraemia (8–10 mEq/L if symptoms start >48 h after surgery), avoiding **osmotic demyelination**.
6. Seizures: benzodiazepines; ICU (intensive care unit).

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy), hysteroscopy.

Diagram to draw: flowchart: absorption → overload + hyponatraemia → cerebral oedema.

Key points to remember:

- Venous pressure 10 mmHg; gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

Rapid Revision Sheet – Practice Set 12 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children (LAQ)

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.

Q2. Lymphatic drainage of testes and principles of designing templates of RPLND (LAQ)

- Right testis → interaortocaval; left → para-aortic; crossover right to left only.
- Boundaries: renal hila, ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve fibres L1–L4 meet at the IMA: unilateral templates spare ejaculation.
- Post-chemotherapy: full bilateral template remains standard.

Q3a. Retroperitoneal fibrosis (SAQ)

- 70% idiopathic; methysergide and lymphoma classic.
- Medially deviated ureters; PET-CT shows activity.
- Prednisone first (Vaglio 2011 beat tamoxifen).
- Bilateral ureterolysis with omental wrap.

Q3b. TIP repair in hypospadias (SAQ)

- Snodgrass 1994: incise, tubularise, heals by epithelialisation.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Q3c. Perineal urethrostomy. (SAQ)

- PU bypasses the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly men.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Q3d. Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options. (SAQ)

- Bulge is most specific; splinting means high grade.
- POP-Q stage II = −1 to +1 cm.
- Sacrocolpopexy most durable; colpocleisis for elderly.

Q3e. Neurotransmitters in erectile function (SAQ)

- nNOS initiates, eNOS maintains; PDE5 terminates.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.

Q3f. TUR Syndrome (SAQ)

- Venous pressure 10 mmHg; gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.